ELECTRO-ACOUSTIC CAN ENHANCE MĀORI MUSIC: A MODEL FOR COMPOSING.

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ABSTRACT

This article examines the possibility of composing electro-acoustic music and integrating traditional Māori waiata/ mōteatea[1] and taonga pūoro[2]. The focus will be on generating a model for composing new works and proving that electro-acoustic can enhance Māori music. The findings of a current PhD research will be highlighted along with 'work in progress' that will give support to this article.

1. INTRODUCTION

As a composer of Māori waiata and a new composer of electro-acoustic music, my principal concern was whether electro-acoustic can enhance Māori music and become an addition to its repertoire. Furthermore can a model for composing new works be generated as a guideline to encourage and develop new composers?

During my PhD research there has been a mixed`rl nana8800 reaction from composers of *waiata* and *taonga pūoro* musicians regarding the fusion of electro-acoustic with Māori music. Rangiiria Hedley (2010), a *taonga pūoro* musician prefers the organic pure sounds of these instruments as a 'stand-alone' genre and regards electro-acoustic as 'weird' due to its inorganic and manipulation of sound via the computer. Also many *taonga pūoro* musicians have a close connection to the instrument whilst playing due to the *'wairua'* the spiritual and emotional aspects of Māori music as explained in Chapter Two *Waiata* and *Tāonga Pūoro.*

On the other-hand Horomona Horo (2010) another taonga pūoro musician enjoys both the pure sounds of the taonga pūoro and experiments with new sounds from these instruments during the recording process. Dr Richard Nunn (2010) a well-known taonga pūoro expert and musician embrace both the organic indigenous sounds of the taonga pūoro but believes that taonga pūoro should be exposed to other 'art-forms'. Nunn has played taonga pūoro on many recordings including an 'electro-acoustic' recording '*Te Hau Kuri*' (2002, NZ Sonic Art III) composed by the late Dr Hirini Melbourne.

In reference to *waiata* and electro-acoustic there have been numerous recordings of *taonga pūoro* and contemporary new *waiata* such as *'Te Ku Te Whe'* (1994), *'Te Hekenga-Ā-Rangi'* (2003), *'Puhake Ki Te Rangi'* (2008) to name but a few. However these are not electro-acoustic recordings but organic sounds of vocal singing with *taonga pūoro* as an accompaniment. I have never heard an electro-acoustic composition that incorporates a traditional Māori *waiata/ mōteatea.* This will be investigated further in the composition of four original music projects in my PhD research. The composition will include two ancient chants of the Tainui and Waikato people, '*Waikato Te Awa'* and *'E Pā Tō Hau.'*

Although electro-acoustic is fairly new to the Māori the debate being, does it have a place in Māori music. This is best argued by Da Cunha (1995):

'Culture is production and not a product, we must be attentative in order to not be deceived; what we must guarantee for the future generations is not the preservation of cultural products, but the preservation of capacity for cultural production'.

Da Cunha argues that indigenous people should treasure their heritage and culture but keep evolving by producing new cultural production. In reference to this article, the suggestion is made that Māori music should adapt and be flexible if Māori as a whole is to survive in the future.

Furthermore, according to Dr Robert Joseph (2007) he expresses the importance of 'reflexivity and adaptability':

'As in the past, Māori have survived dramatic changes of colonisation, urbanisation and now globalisation, individually and collectively, by deploying their capacity for adaption; on the one hand modifying traditional forms to serve new functions and on the other creatively adapting introduced forms to their own ends, transforming both in the process'.

[1] **Waiata** is the generic term for all Māori songs. Mōteatea refers to laments and poetic prose of Māori composers.
[2] **Taonga pūoro** refers to all traditional Māori instruments.

The Māori have survived into the 21st century and they have brought forth their tribal heritage, genealogy, traditional belief systems, cosmology, customs and traditions, language, myths and legends, guardianship *kaitiakitanga*, and on the 'creative side' all the arts including *waiata, taonga pūoro* and the possibility of including *electro-acoustic*.

2. WAIATA AND TAONGA PŪORO

Waiata and *taonga pūoro* are part of the traditional music of the Māori people. Tiramorehu (1849) stated:

'Kei te Pō te timatatanga o te waiatatanga mai a te Atua. Ko te Ao, ko te Ao mārama, ko te Ao tūroa'

Translated, 'it was in the night that the Gods sang the world into existence. From the world of light into the world of music.

According to McLean (1996, 1999) the traditional Māori instruments consisted of idiophone (percussion instruments) and aerophones (wind instruments). Membrnophones (drums) were absent and there was only one chordophone (stringed instrument) represented by the *Kū* originating from the South Island of New Zealand.

Tiramorehu (1849) states that Māori music is connected with the Gods who in turn provide the sounds that are imitated by the musicians who sing *waiata/ mōteatea* and play *taonga pūoro*. Flintoff (2004) states that *taonga pūoro* instruments are part of the family of traditional Māori instruments that is categorized under Māori gods such as *Ranginui* the tunes; *Papatūānuku* the rhythms; *Hine-Pū-Te-Hue* the gourds; *Tānemahuta* father of song makers; *Tangaroa* the father of shell instruments; *Tāwhirimātea* the wind voices, and *Raukatauri* goddess of flutes.

Since the 1980s there has been a revival of both the *waiata/ mōteatea* and *taonga pūoro* traditions in recent times. Extensive collections by Ngata & Jones (2004, 2005, 2006, 2007) in the published series *Ngā Mōteatea 1, 2, 3, 4* and McLean Waiata Collection and published books, *Māori Chant:a study of ethnomusicology (1965)*, *Māori Music (1996), Traditional Songs of the Māori (1975)* has inspired the re-learning of ancient chants that are performed on the *marae*, tribal workshops and at regional and national kapa haka competitions.

Whalley (2005) mentions the cultural revival of Traditional New Zealand Māori instruments led by pioneer and performer/ composers Dr Hirini Melbourne and Dr Richard Nunn. Whalley also mentions live performance, improvisation, crossover work and collaboration between the composer and performer of *waiata* and *taonga pūoro* in recordings such as *Te Ku Te Whe* (1994) and *Te Hekenga-Ā-Rangi (2003)*. Both these recordings were a collaboration between Dr Hirini Melbourne and Dr Richard Nunn. The albums featured live recordings of *taonga pūoro* and *waiata.*

Other works featured Gillian Whitehead (1998), Helen Fisher (2000) and Martin Lodge (2004) who created a number of acoustic 'art music' thus giving rise to 'cross over' works. Whalley (2005) states;

'Although outside the digital domain, this work is touched on here to provide an insight into methods of working with composers both within and beyond the Māori music tradition.

To date the recording of *Te Hau Kuri (Dog's Breath)* composed by Dr Hirini Melbourne and featured on New Zealand Sonic Art III (2002) provides a fusion of electro-acoustic sound with *taonga pūoro* being the main instrument. The sounds of the nine traditional Māori instruments were deliberately chosen and arranged to retell the story of the Okiwa wind musically. This particular recording gives a true organic sound of *taonga pūoro* within an electro-acoustic setting.

3. A MODEL FOR COMPOSING

Analysing data from literature, personal communications with composers (of waiata, taonga pūoro and electro-acoustic), and examples of works gave rise to formulating a model for composing, thus the **Harakeke and Tūī** model. Although this is only one model for composing it takes into consideration important stages of composing original music; the pre-requisite stage, the production stage and finally the product stage.

3.1. The Harakeke and Tūī model

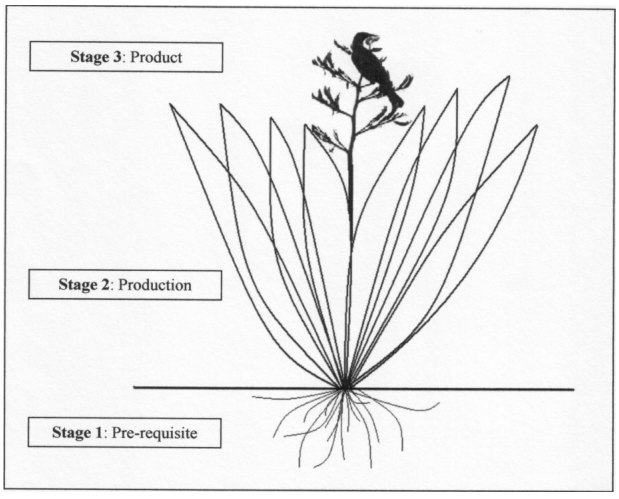


Figure 1. The Harakeke and Tūī model (Rollo, 2010)

Key

Stage One in the model is the pre-requisite for the composition of waiata, taonga pūoro and electro-acoustic. Each genre requires different knowledge and

skills in the composition process before progressing on to stage two and three.

Stage Two is 'the hands on' production of waiata, taonga pūoro and electro-acoustic composition. Each genre will be treated differently in the composition process and requires artistic creativity and an extensive knowledge of music production before progressing on to stage three.

Stage Three is the final product and digital recording of the waiata, taonga pūoro and electro-acoustic composition. The final outcome is the recording (music recording studio or mobile recording studio) and live performance of each composition to showcase the original compositions and to prove the validity of the new generated model for composing.

During stage three there will be an evaluation of each composition by invited participants (composers, performers and listeners) with the intention of improving the quality of the new works. Even after the final digital recording, there will be on going critical review which is eminent in creative arts.

3.2. The Philosophy of the Harakeke and Tūī model

Figure 1 represents a Māori perspective of the three main stages in the composition of waiata, taonga pūoro and electro-acoustic music.

Stage One is represented by the roots of the *harakeke*, Phorium tenax – New Zealand flax. The flax cannot survive without soil, water, the sun, its nutrients and roots that gives sustenance to the *harakeke* plant.

In regards to the composition process, the pre-requisites, the knowledge and skills of the composer are vital.

Stage Two is represented by the *rau* or flax leaves. Each leaf represents the important aspects of composition (the production) or 'hands on' approach. This includes the theme, organic sound samples, synthesized sound samples, live recordings, the arrangement, sections of the composition, the editing and so on.

Stage Three is represented by the *kōrari* (the flower stem of the flax) and the *tūī* bird (Prosthemadera-novaeseelandiae) a native New Zealand bird which eats the nectars of the flowers on the *kōrari* stem. In Māoridom the *tūī* is known for its sweet chirping singing voice and portrays the sound of the final composition hence the performance and final digital recording.

'Ka tangi te tūī, te kanakana te hae'
When the tūī sings, the jealous eyes in on the watch

4. MUSIC TECHNOLOGY AND COMPOSITION

Music technology has always played a big part in electro-acoustic music from the 'reel to reel' recorder, the phonograph recording, the 'tape cassette' recorder, sampling machines, Telharmonium (1897), Chamberlain (1946), Mellotron (1960s), Birotron (late 1970s), Optigan (1971), Optigan (1971), EMS Synthi AKS (1972), Orchestron (1975), Synclavier (1975), Fairlight CMI (1978), Kraakdoos, DX7 (1983), UPIC to the more sophisticated Macintosh e Mac, G4-700 combo, Roland PC-300 Keyboard, Steinburg Cubase AV, and supporting software programs such as Protools, Metasynth syntheziser and Kontakt sampler.

In reference to the history of electro-acoustic/ electronic music Collins and D'Escriván (2007) gives an overview and the importance of electrical technology:

'From musique concrete to audio-visual sampling, from eletronische music to minimal techno, from the Telharmonium to the laptop, electrical technology has facilitated more than a century of original music, spawning a multitude of new styles, instruments and methods'.

How can technology and the composition process support the fusion of electro-acoustic and Māori music focussing on the introduction of *waiata* and *taonga pūoro*? This will be further investigated during the composition of the four music works in the PhD research. The **Harakeke and Tūī** model provides a starting point in accumulating ideas and planning the new composition and, as technology advances into the 21st century and beyond will bring the music to reality.

The exciting part of the production stage is to experiment with live recordings of *waiata* and *taonga pūoro* and to decide whether to input the organic sounds as they are or to manipulate and invent a new sound, keeping in mind cultural considerations (Māori context). The composer must also give thought to where the *waiata* and *taonga pūoro* should be placed in the composition to highlight the motif.

5. WORK IN PROGRESS

As part of my PhD research, there will be four music works to test the models for composing *waiata, taonga pūoro,* and *electro-acoustic.* This is the exciting and creative part of the research. I will be working collaboratively with composers of the fore mentioned genres, musicians and engineers to produce the original music using the **Harakeke and Tūī** model. Although the work is still in progress I will give you some background information into the concept and process.

5.1. The Planning Stage

Figure 2 gives an outline of the overall theme of the four original music compositions which includes:

- One *taonga pūoro*[3] music composition.
- One *waiata tawhito*[4] music composition

[3] **Taonga pūoro** refers to all Māori traditional instruments. In reference to this article the composition refers to a live performance recorded.

[4] **Waiata tawhito** refers to ancient chants and song of the Māori.

- One *waiata hou* music[5] composition
- One *electro-acoustic*[6] music composition

The four music compositions will be integrated together to give a dramatic account of the origins of the Waikato river, the establishment of the *Kīngitanga* Māori king movement, a lament to the late Māori queen Te Arikinui Te Ātairangikaahu, historical account of the Waikato wars and the battle at Rangiriri *pā*[7] and finally the celebration of Kīngi Tuheitia, the sixth Māori king. *Waiata, taonga pūoro* and *electro-acoustic* will feature in this music project.

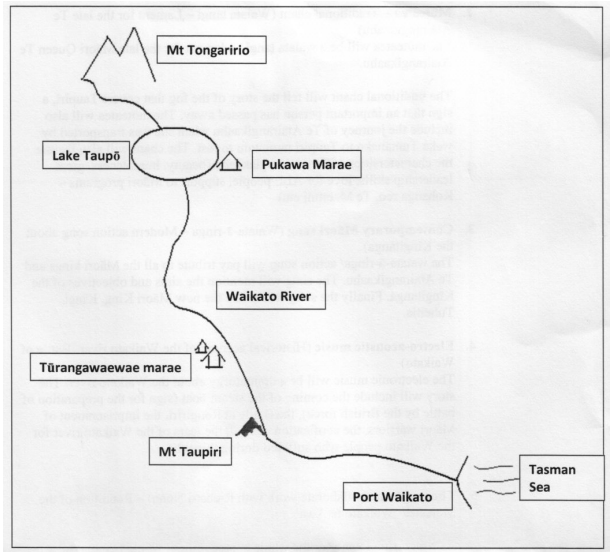


Figure 2. The Planning of the Four music compositions

5.2. The Four Music Composition

The reoccurring theme that will unite the music together is *Waikato Te Awa* a traditional Māori chant. This *waiata* celebrates the Waikato River the landscape that embraces the four music compositions as detailed in Figure 2. The other *waiata* that will feature in the composition is *E Pā Tō Hau* a lament.

Composition One is a *taonga pūoro* music that will tell the story of the origins of the Waikato River and the meeting at Pukawa marae that established the *Kīngitanga* Māori king movement. This will be a collaborative work with Horomona Horo, a *taonga pūoro* musician.

Composition Two is a *waiata/ mōteatea* a lament to the late Māori queen Te Arikinui Te Ātairangikaahu. The lament mourns her death and her travels on the *Tumanako* canoe to her final resting place at Taupiri mountain. Māori lyrics will be written and a melody line sorted. The form of the *waiata/ mōteatea* is a *waiata tangi*[8] and will follow traditional Māori music aesthetic.

Composition Three is an *electro-acoustic* giving a historical account of the Waikato war and the battle at Rangiriri pā. A period of unrest between the colonial army and the Waikato tribes.

Composition Four is a *waiata hou* a contemporary Māori action song that acknowledges all the Māori kings and Te Arikinui Te Ātairangikaahu. This song celebrates Kīngi Tuheitia, the sixth Māori king.

The purpose of the four music compositions is to test the new generated models for composing *waiata, taonga pūoro* and *electro-acoustic*, a guideline to develop new composers in the future. After the compositions are completed, invited composers, performers and listeners will evaluate the music and give feedback. In turn, the compositions will be edited and re-worked before a final digital recording is produced. There will also be a DVD recording of the *waiata* and *taonga pūoro* performances.

6. ELECTRO-ACOUSTIC AND MAORI MUSIC

Why not?

Composers and performers of electro-acoustic music have already indulged into creating new works individually or in collaboration with others. There has been new works that have incorporated other music forms and themes such as the recording *Kasumi* (2002, New Zealand Sonic Arts III) composed by Ian Whalley.

This particular composition includes a Japanese text from *Manyoshu* number 4290 by Otomo no Yakamochi (23 February 753) and the integration of *taonga pūoro.* The short instrumental samples used in this work are from seven New Zealand Māori instruments. In *Kasumi* (for instrument (s)/ voice and tape) the original sounds provided the basis for the electro-acoustic manipulation and integration of the work. Based on the text, the composition is constructed using parallel narratives like a folk tale, juxtaposing simple binary elements (2002, New Zealand Sonic Arts III).

The cultural/ environmental exploration of a New Zealand musical voice has seen the revival of both *waiata/ mōteatea* and *taonga pūoro.* This extended the Māori instrumental tradition through the application of digital music production techniques (Whalley, 2005), thus creating new works. The PhD research involves the composition of the four music compositions as detailed in 5.2. This will explore further organic and manipulated sounds of voice (*waiata)* and traditional instruments (*taonga pūoro).*

[5] **Waiata hou** refers to contemporary Māori songs with Western and other world influences.

[6] **Electro-acoustic** is electronic music.

[7] **Pā** is a fortified Māori village or settlement.

[8] **Waiata tangi** is a lament to the dead.

Māori music is evolving and Māori composers, performers and audience/ listeners have been exposed to many music genres including electro-acoustic. Although the revival of traditional *waiata* and the making and playing of *taonga pūoro* is eminent today, Māori music will explore new avenues, create new sounds and add to the vast repertoire of New Zealand music.

Electro-acoustic can enhance Māori music and the PhD research will prove that the models for composing is practical and the new compositions support the idea that *waiata* and *taonga pūoro* are important Māori art-forms that can be fused with Western music such as electro-acoustic, or any other forms of music for that matter.

7. CONCLUSION

This article examines the intellectual and creative aspects of composing new electro-acoustic music and combining *waiata/ mōteatea* and/ or *taonga pūoro* in to the soundscape/ abstract composition.

The ongoing current PhD research will generate a model for composing *waiata, taonga pūoro* and *electro-acoustic* with the aim of cross pollinating musical ideas in a cross-cultural environment. This model will be tested with four 'creative' music compositions not only to validate the model for composing but to support that electro-acoustic can enhance Māori music, and the production of new sounds and new works.

The Māori proverb *'He taurangi te hau, he hurihuri te ao'* supports the idea that *waiata, taonga pūoro* and *electro-acoustic* music is evolving and undergoing many changes, thus providing new concepts, theories, music practices, methodologies and technologies to enhance compositions for the future generation to employ and create new works of art.

8. ACKNOWLEDGEMENT

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